

STATE CYBER CELL — MOM & AUDIT LEDGER PLATFORM

Complete Technical Feature Overview & Live Demonstration Guide

System Name	State Cyber Cell Minutes of Meeting (MoM) & Cryptographic Audit System
Architecture	Hybrid Dual-Engine: React Vite (Port 5173) + Express Node (Port 5000) + Python ML (Port 8000)
Security & Compliance	SHA-256 Cryptographic Hash Chain Ledger + Role-Based Access Control (RBAC)
Deployment	Zero-Admin Portable Python (`setup.bat` / `setup.sh`) + 1-Click Vercel Serverless Ready

1. CORE FEATURE & CAPABILITY MATRIX

Feature Component	Technical Capabilities & Demonstration Behavior
1. Live Audio Ingestion & STT	Accepts local audio file uploads (.wav, .mp3) OR records live via microphone. Displays a live accumulative Web Speech API transcript text stream with word counter and immediate hardware microphone track release (<code>stream.getTracks().forEach(t => t.stop())</code>).
2. Smart AI Case Title Engine	Semantic NLP classifier analyzes full transcript content to identify crime category (LockBit Ransomware, Deepfake Extortion, SIM Swap, Crypto Seizure, Phishing) and extracts spoken FIR/Ticket numbers (<code>[FIR-2026-8890]</code>) into authoritative titles.
3. PII Anonymization & Role Gating	Detects sensitive entities (<code>`FIR_ID`</code> , <code>`BADGE_ID`</code> , <code>`CYBER_TICKET`</code> , <code>`PHONE_NUMBER`</code>). Interactive click-to-toggle mask/unmask pills with Auditor clearance level locking (<code>`AUDITOR`</code> role restricted from viewing unmasked PII).
4. MoM Editor & Action Items Matrix	Structured MoM generator producing Agenda Topics, Key Decisions, and Editable Action Items Matrix (Task, Owner, Deadline, Status). Direct inline case title editing (<code>=■</code>) with immediate state update.
5. SHA-256 Cryptographic Audit Ledger	Immutable audit log (<code>`hash = SHA-256(data + prevHash)`</code>) anchored by a Genesis block. Dual-View toggle (<code>■ Human-Readable View`</code> vs <code>■■■ SHA-256 Hash Chain View`</code>). Real-time event logging for audio play, view mode switch, PII unmask, title edit, and sign-off.
6. State Police PDF Exporter	Generates official State Police print-ready PDF reports (<code>`PdfReportModal.jsx`</code>) formatted to law enforcement document standards for court submission.
7. Universal Cross-Platform Setup	Automated setup scripts (<code>`setup.bat`</code> for Windows, <code>`setup.sh`</code> for Mac/Linux) that detect system Python or download Portable Python 3.11 with zero admin prompt required.

2. STEP-BY-STEP LIVE DEMONSTRATION WALKTHROUGH

Follow these exact steps during your presentation to showcase the full end-to-end flow:

Step 1: Launch Application

Run ``npm start`` in terminal. Open browser at ``http://localhost:5173/``. Point out the sharp, boxy Police Command Terminal interface.

Step 2: Dynamic Role Clearance Switch

Use the top header dropdown to switch between **INVESTIGATOR**, **ADMIN**, and **AUDITOR**. Show how the **AUDITOR** role automatically masks raw transcripts and restricts PII unmasking.

Step 3: Live Microphone Recording & Audio Ingestion

Click **Process Meeting Audio** → **Record Live via Microphone**. Speak: *'Inspector Deshmukh investigating blackmail attempt via fake profile targeting victim under case FIR 2026 8890.'* Click **Stop Recording & Release Mic** → **Process Audio File**.

Step 4: Smart AI Title & Extraction Verification

Show how the AI instantly extracts the title: **Deepfake & Cyber Extortion Threat (Targeting Student Victim)** **[FIR-2026-8890]** and populates the Agenda, Decisions, and Action Items Matrix.

Step 5: Interactive PII Unmasking & Audit Log

Click on the redacted PII tag ``[FIR_ID: .....]``. Watch it reveal ``FIR-2026-8890``. Scroll down to the Audit Ledger and show how the event was cryptographically logged into the SHA-256 hash chain!

Step 6: Cryptographic Hash Chain Verification

Click the toggle **SHA-256 Hash Chain View**. Show your friend the mathematical proof (``hash = SHA-256(data + prevHash)``) preventing any record tampering.

Step 7: Official PDF Export

Click **Export PDF Summary**. Showcase the official State Police print layout formatted for court and command review.

3. SINGLE COMMAND LAUNCH FOR ANY PC

To run on any Windows computer:

```
setup.bat
npm start
```